

Top Insights from Chat

Agent vs workflow · Tools · Memory · Loop control · Semantic routing

The chat ran hot for all three hours, with Krishna answering in the thread while Dhaval and Sid taught. What follows is the substance of it: the questions useful to everyone, grouped by theme and attributed to whoever asked, with the answer where one was given. Housekeeping and setup chatter have been left out.

Key Insight (from venkat)

"Krishna is acting as agent in today's class." A joke, but it caught something real. A large share of the learning in this session happened in the chat thread rather than on the slides.

1. What Makes Something an Agent

Definitional questions dominated the first hour, and several people pushed back on the definition rather than accepting it, which is what the mentors were asking for.

Asked by	Question	Answer or takeaway
Sharath S	How is "action" different from "tool calling"?	An action is a task you want performed. Tool calling is invoking an API or function. One action may need several tool calls. Dhaval's carpenter: "build a chair" is the action, the hammer and saw are the tool calls.
Bharath Kumar	Is it an agent only when it makes the payment, or is adding to the cart enough?	Krishna: neither is the test. Both are just tool calls. What makes it an agent is pursuing your goal autonomously.
Renganathan M	Placing an order also uses tools. Is that not correct?	Krishna: yes, it will use APIs as tools. The distinction is the goal, not the mechanism.
Ankit Garg	How do RAG and AI agents fit together?	Krishna: RAG answers questions by retrieving information. An agent decides what actions to take and which tool calls those need. An agent may use RAG when it needs that information.

Asked by	Question	Answer or takeaway
Anupam Saraswat	Still not convinced this is an agent, since ChatGPT does the same and we said that was not one.	Sid: the model inside ChatGPT is not an agent, and early versions could not use tools at all. Once tools, search and memory arrived in early 2023 it became agentic. Agency has levels, and what you built today is a simple first degree agent.
Mohan	Here the LLM is the orchestrator, right?	Yes. The LLM is the brain, with tools, knowledge and memory attached around it.
sunil	When does an agent become agentic?	When it takes autonomous action toward a goal. Calling a tool alone is a tool augmented chatbot.

Key Insight (from Sandip)

"Action equals act to perform a given user goal." Sid marked this one correct in the thread. If you remember one line from the first hour, this is a good candidate.

2. Guardrails and Safety

Asked by	Question	Answer or takeaway
Goutham	The rope is the guardrail?	Yes, and he named it before Dhaval got there. The cow is autonomous inside the circle the rope allows, and cannot reach the road or the neighbour's crop.
Anupam Saraswat	Would grounding and a perimeter fence also count as guardrails?	Confirmed. Any boundary on what the agent may do counts.
Venkateswara Rao K	How do we define guardrails?	Krishna: coming soon. Dhaval confirmed a full separate session is planned.
Anurag Pandey	The OpenAI and Hugging Face incident. It went outside the sandbox.	Anurag surfaced the case Dhaval was fishing for: a model trained for cybersecurity work broke its guardrails and attacked Hugging Face. The lesson: even defined guardrails can break, so making them solid is your job.

3. Tools, Decorators and Docstrings

The busiest technical thread of the session, and the answers most worth keeping.

Asked by	Question	Answer or takeaway
Anurag Pandey	What does the tool decorator do? What changes inside the function?	Krishna: nothing changes inside it. It registers the function so the agent can discover and call it.
im	By registering functions as tools, does LangChain start executing them automatically?	Krishna: no. The decorator only makes the function available. It runs when the LLM decides to call it.
Renganathan M	Only callables can be made into tools?	Yes.
raghav bansal	Do we decorate each tool individually?	Yes, every function you want exposed.
umesh	How do we restrict the model to only the tools we configured?	Krishna: that is what bind_tools does. The model can only choose from the list you bind.
Nltesh	Who decides which tool to call, the LLM or LangChain?	The LLM. LangChain only registers tools and passes their descriptions into the system prompt.
Renganathan M	What if our tool is written in Java or C?	Krishna: the agent does not care about the language. It needs a callable interface, usually a REST API.
Bharath Kumar	If I swap the docstrings between two tools, will the agent get confused?	Krishna: yes. The docstring is how the model knows what a tool does, so a wrong one misleads it directly.
Bharath Kumar	If two tools can do the same thing, how does the agent choose?	Sid: common, and generally not good practice. Make each docstring say what that tool is best for. In practice it may try one, judge the result thin, and try the other. Weak models get confused, strong ones work it out.
im	A function is hardcoded, so how can it answer something outside its scope?	Krishna: that is exactly the split. The LLM reasons and selects. The function is only the deterministic execution part.

Key Insight (from CHAITHANYA)

"Now colleagues are LLMs." Written when Sid asked what a docstring is for these days. Docstrings used to be a courtesy to the developer next to you. They are now the interface through which a model decides whether and how to call your function, which is why a sloppy one breaks an agent.

4. Why Model Behaviour Varies

A cluster of questions came from people running the same code and getting different results.

Asked by	Question	Answer or takeaway
Aditi Reddy, Nitesh	How is it converting my question into a ticker symbol, or spotting that Google is a company?	The LLM is pretrained on enough text to know NVIDIA trades as NVDA and Google as GOOGL. That knowledge is intrinsic, not looked up.
supreet kaur, im	Why did some people get no result on the first query, and why is each response different?	Krishna: LLMs are probabilistic. The model may pick the wrong company or the wrong tool, and the same prompt can take a different path.
Ashu Swami	In "how is Dhaval Patel's company doing?", what is the ticker in that prompt?	Krishna: there is none. The model infers it and passes it as an argument. Codebasics is not listed, so there is nothing to find.
Amit Ashok Kulkarni	Shared a 404 from Yahoo for ticker AIRTEL, all fields null.	A useful failure. AIRTEL is not a valid Yahoo symbol, so the model guessed wrong and the tool returned nothing.
Abdoul	So model APIs have no browsing feature like the playground?	Sid: correct. Call the API for any current model and it will only generate text. Browsing is a product feature, not an API one.
Gourav	I see no credentials anywhere, so how is the API being called?	yfinance uses Yahoo's public endpoints under a free limit, so no key is needed. Sid's follow up was pointed: one Google search would have answered it.
Aditya Bharadwaj	SERP API now makes a phone number mandatory.	Sid: recent, and likely a response to people creating multiple free accounts. If you would rather not, use Tavily or another search library.

5. Memory, State and Sessions

Asked by	Question	Answer or takeaway
Harendra, Ankit Garg	Which session ID do we pass, and how do we make memory persistent per user?	Krishna: on a new chat, generate a unique thread ID (a UUID), store it, and reuse it for that session. Sid: this is an ordinary backend decision, no different from assigning post IDs. The AI layer only needs the thread ID handed to it.

Asked by	Question	Answer or takeaway
priya, Ankit Garg	Is SQLite used for checkpoints in real production systems?	Krishna: generally not. In production it is PostgreSQL, Redis or MongoDB.
satyam	How do we specify short term versus long term memory?	Krishna: a memory saver for short term, a persistent database for long term.
Sharath S	Even with a persistent checkpointer we pass a thread ID, so it is still session level. How do we get long term?	Sid: correct, and long term memory is a later session. For now, experiment with short term.
Anurag Pandey	How is the database schema decided?	Sid: you do not decide it. LangChain manages its own schema, the way Keycloak or Celery create their internal tables once you hand them a connection.
Gourav	I ran it but cannot see how the memory actually works.	Sid's recap: it checks the thread ID, writes messages to a table keyed by that thread, and on the next question fetches that thread's messages into what it sends the model.
satyam	Can we cache so we do not call the LLM every time for similar questions?	Sid: yes, semantic caching, covered in the optimisation session.

Key Insight (from sunil)

"It's like RAM, long term is like hard disk." The clearest short explanation of the memory split anyone offered, and worth borrowing when you explain it to someone else.

6. Controlling the ReAct Loop

Once the agent was running the questions turned practical, and mostly financial: how do you stop it burning tokens?

Asked by	Question	Answer or takeaway
Rakkappan, Aditya Bharadwaj	How does the ReAct loop come to an end, and how does it know?	Sid: when the agent stops making tool calls. Once it judges it has done enough it issues no further calls and generates the final response, which closes the loop.

Asked by	Question	Answer or takeaway
Pujan, Sandip	Can the loop run forever? How do we stop it after three or four iterations before it burns tokens?	Sid: set a maximum iteration count. Cap it rather than trusting the model to be frugal.
Guruprasadh Swaminathan, supreet kaur	Does max retries also cover the ReAct loop?	Krishna: yes, that parameter caps how long the loop runs.
Himesh Sarkari	When we ask which has better growth prospects, will it still call the tools and follow the format?	Krishna: if the comparison needs fresh data, yes. Sid: it depends on the model and the moment. In the live run the agent already held both reports, so it compared them and called nothing, which is itself agency.
Aditi Reddy	Is there a cap on the number of tools?	Not answered live. Connects to Sandip's hundred tools question below.

7. Semantic Routing

Routing produced more chat questions than any other topic. The threshold and accuracy discussion was driven almost entirely from the chat.

7.1 Building and training the router

Asked by	Question	Answer or takeaway
Pujan	Do we pass so many questions every time? Is there a generic way?	Sid: no. Utterances and threshold tuning are a one time build step. At inference you pass only the user's question.
Rakkappan	What are X train and Y train?	Sid: terminology from machine learning. X is the input, here the question. Y is the expected output, here the correct route.
Karthik T	Are utterances also training data?	Yes. Utterances are the example queries embedded and stored against each route.
Goutham	Why 0.5, and how does that number affect the answers?	Krishna: it is the library default and can be changed. Lower accepts more queries as matches, so more misrouting. Higher accepts only very close matches.

Asked by	Question	Answer or takeaway
Goutham	Did we split the dataset? Do we not need 70:30?	Sid: no split needed here, because a separate test dataset was written from scratch.
Sayee Sree Ram	How is accuracy calculated?	Sid: out of the 35 questions, the percentage routed correctly. Nothing more.
Amit Patil	Where does the router persist its training for next use?	Sid: give it a vector database or other persistent storage and it will store there, ready to use directly.
Mugdha	Can we provide files as data in the route config?	Yes. Define routes as JSON and load from file. The docs cover save and load plus Postgres, Pinecone and Qdrant indexes.
im	Do we need an LLM for the first time embedding?	Krishna: no, an embedding model. No LLM is involved in routing, which is the whole cost argument.
Sandip	Is it allowed to fine tune embedding models? That is repetitive work.	Sid: yes, and it is not repetitive. Fine tune once and reuse.

7.2 Accuracy, overlap and edge cases

Asked by	Question	Answer or takeaway
im	How do we achieve 100 per cent accuracy?	Sid: nobody knows. In production he would generate a large dataset, fine tune the embedding model, write fresh utterances, and route on the fine tuned model. Realistic expectation is 80 to 90 per cent.
Shubradip Ghosh	What percentage of data is needed for a training dataset?	Sid: depends on the use case. He then challenged the assumption that these routes were simple: stock analysis, market news and general finance overlap heavily, so the borders between their embeddings are genuinely blurred.
Pujan, Anurag Pandey	What if no route matches, or we failed to cover a category?	Krishna: there is a default route that such queries get directed to.
Ankit Gupta	How do we decide the number of routers?	Krishna: one per distinct business intent or workflow, not one per API or tool.
Sandip	With 100 tools, do we prepare training data for each tool?	Sid: no. Cluster tools into the agents or pipelines that use them and build one route per pipeline.

Asked by	Question	Answer or takeaway
Pawan, satyam	Can two routes tie? What if two agents can handle the same request?	Krishna: the one with higher confidence wins. Sid noted the library resolves ties internally.
Guruprasadh Swaminathan	If a query combines all three route intents, which wins?	Answered from the cohort by Srinivas Vadla: rare, and when it happens you define which route takes priority.
satyam	What if the router picks the wrong agent?	Srinivas Vadla: the wrong tool accepting the query produces irrelevant results, which is why threshold tuning matters.
Anurag Pandey	Could we use knowledge distillation for a small classifier? Overkill?	Sid: no distillation needed. You can fine tune a small model for classification directly. Covered in the fine tuning session.

7.3 How routing relates to everything else

Asked by	Question	Answer or takeaway
Renganathan M	How is semantic routing related to the agents from the first half?	Krishna: routing decides which agent or workflow handles the request. The agent then decides which tools to use inside it.
Aditi Reddy	Is tool calling not also a kind of semantic routing?	Sid: the LLM decides which tool to call. Semantic routing decides which agent or pipeline handles the request at all. Different layer.
supreet kaur	Is the difference from RAG that there we saved embeddings in a vector store and here we did not?	Sid: no, you can save these too. The difference is purpose. In RAG the embeddings feed retrieval for generation. Here they do intent classification.
Nltesh	Is it storing to a disk file?	Krishna: yes, that is what auto sync local does. Without it, embeddings regenerate on every application start.
satyam	An LLM can also act as a router, it just costs more, right?	Confirmed. That cost is exactly why the embedding approach exists.
Aditi Reddy	Coding agents specify one model. Where does routing fit there?	Sid: in auto mode the coding agent decides which model to use per task. That is complexity based routing doing route classification.

Key Insight (from akhil)

Answering Aditi in the thread: "We use semantic routing when we need to handle multiple tool calls across different scenarios. Calling all tools simultaneously is both time consuming and expensive in terms of token costs. Semantic routing solves this by analysing the meaning of the user's query and matching it to the correct tool." Several people endorsed it, and it beats most textbook definitions.

8. Learning, Practice and Career

Asked by	Question	Answer or takeaway
Anurag Pandey	Coding agents build end to end, so where do we stop them and what should we still do ourselves?	Sid's workflow: write the skeleton yourself (folders, modules, core functions), leave the rest as stubs with comments, have the agent generate a PRD, verify it in depth, then ask for phased tasks and implement phase by phase, reviewing each phase first. Hand over everything in one prompt and you get more output than you can review.
hema	We understand the concepts, but writing code for even a simple pipeline is very difficult.	Sid: not enough practice. Nobody could code a week after learning to program. The mistake is jumping from concepts straight to the pipeline. Draw the components on paper first.
Prasanna	No prior coding experience. Any guidance for people like me?	Sid: use an AI agent to write the code, but read everything it produces and work to understand it. Go through the Python module alongside.
im	Embedding models are trained with ML. Do I need to learn ML too?	Sid: for the application layer, very little maths, and what is needed will be covered. Research is different and needs calculus, linear algebra and statistics.
Sayee Sree Ram	What is the path into research? I have 8 years as a software developer.	Sid: learn the maths, read papers in order from AlexNet through the pre transformer attention work, Attention Is All You Need, then OpenAI, DeepSeek and Llama. Learn PyTorch, implement your own transformer, rent a GPU, improve a component, publish. A previous cohort member did exactly this.
Goutham	In a fast changing field, what should the core focus be: maths, code, or tools?	Sid: the concepts and techniques, not the code. Semantic embedding has appeared in every lecture so far. Dhaval's version: LangChain community was taught last cohort and is deprecated in this one, but the principles did not move.

Asked by	Question	Answer or takeaway
Gourav	How do we set this up on a local machine?	Sid: hand the notebook to Claude and ask it to convert it into a modular Python project. Only the packaging changes. akhil added that Kiro or Cursor free credits replicate the Colab locally.
Amit Ashok Kulkarni	Can an agent generate images or video? Which model?	Sid: Google's Omni model and Nano Banana, GPT Image, or the open source Flux models.

Key Insight (from Karthik Prabhu)

"We should focus more on the core concepts than the code, as frameworks such as LangChain have extensive documentation, which we can refer to while we get our hands dirty." Dhaval's reply in the thread was one word: "exactly".

9. Standout Contributions from the Cohort

Who	What they contributed
Srikanth Adilapuram, Annu	Both completed the JSON output homework live and posted clean structured briefs. Sid asked for the code.
Sandip	Posted a working message serialiser for printing the agent trace as JSON, on top of several of the sharpest architecture questions of the day.
Srinivas Vadla	Fielded two routing edge cases for the room: define a priority route when intents overlap, and expect irrelevant results when the wrong route wins.
akhil	Explained semantic routing versus tool calling in the thread, and suggested Kiro or Cursor as a local alternative to Colab.
Anupam Saraswat	Framed the difference as "api call is a get, action is a post", and kept pressing on the ChatGPT definition until it was properly resolved.
Augusto	"It makes more sense now to me how ChatGPT and others work regarding the model you choose." A good marker of when the routing architecture landed.
Aditya Bharadwaj	"Now we are addicted to Claude Code, earlier we wrote everything line by line." Also flagged the SERP API phone number change for everyone.
Krishna A, Prajvan	Two good one line definitions of a router: intelligent API direction, and "like a receptionist".

10. Tools the Cohort Is Using

Category	Named in chat
Coding agents	Claude Code (most cited, and Dhaval's example of a high agency agent), GitHub Copilot, Cursor, Kiro, Codex, Devin, Antigravity, Replit.
Autonomous agents	OpenClaw and Hermes at the high agency end, UiPath and Make on the automation side.
Infrastructure people already know	Celery, Kafka and Keycloak, which came up when Sid used them to explain that LangChain manages its own schema. Several confirmed they had used all three.
What people plan to build	An AI news agent (Bharath Kumar), an email manager (Adil), an AI job hunter (Amit Ashok Kulkarni, Aditya Bharadwaj), SQL database agents (Murthy Rajender), and a document Q and A RAG agent (Sushma).